

Media Archaeology of Inscribed Sound: Optical Reconstruction and Artifact Isolation in Fragile Historical Recordings

Author: Dr. Yuna Takahashi • Co-authors: Kaveh Danesh, Marcus Brody

Published: 2023-09-12 • Accession: ESSAY-2023-02 • Canonical: <https://zazieinstitute.org/monographs/essay-2023-02>

ABSTRACT

Fragile historical sound media—including nineteenth-century wax cylinders, early acetate discs, and experimental lacquer surfaces—frequently suffer from mechanical degradation that prevents stylus playback. This paper documents an optical scanning methodology that extracts high-fidelity audio without physical contact. Using dual-laser triangulation and confocal profilometry, we record high-resolution surface topographies, isolate intentional groove modulations from mold spore pitting and surface fissures, and reconstruct historic audio with unprecedented clarity.

KEY THEOREMS & BOUNDARY EQUATIONS

Formulation [1.1]

Principle of Non-Contact Groove Reconstruction: Optical triangulation resolves sub-micron lateral groove wall excursions at 192 kHz sample equivalent without imposing mechanical stylus drag.

Formulation [2.1]

Artifact Decoupling Invariance: Microscopic surface damage exhibits steep vertical edge profiles distinct from the continuous sinusoidal curvature of engraved acoustic signals, allowing algorithmic artifact separation.

Formulation [3.1]

Vitrified Sound Extraction: Even on fragmented ceramic and glass surfaces containing accidental historical vibrations, optical interferometry recovers recognizable frequency traces.

1. The Vulnerability of Early Sound Carriers

Mechanical playback relies on a diamond or sapphire stylus exerting several grams of downward force onto delicate grooves. For aged cellulose nitrate, beeswax, and vulcanite records, every playback pass causes irreversible physical loss. Optical reproduction decouples the recovery of sound from physical wear, opening fragile historical collections to safe acoustic study.

2. Optical Laser Profilometry Architecture

Our apparatus mounts fragile media on an air-bearing rotary stage with sub-micron axial runout. A 405 nm blue semiconductor laser projects an optical knife-edge across the groove wall, while a high-speed linear CMOS sensor captures reflected diffraction patterns at 200,000 frames per second. By measuring lateral displacement across both groove walls independently, we generate stereo audio files directly from geometric surface data.

3. Computational Artifact Isolation

The raw surface scan contains both acoustic modulation and physical flaws: dust fibers, cracks, and mold gouges. We developed an algorithmic filter that tracks groove wall trajectory curvature. Because intentional acoustic recordings obey strict acceleration limits governed by original recording horns, sudden sharp step discontinuities can be cleanly flagged and bridged without dulling musical transients.

4. Case Studies and Archival Restorations

We present restorations of previously unplayable cylinders from 1898 to 1912, including historic field recordings of indigenous oral poetry and early laboratory tuning fork tests. Recovered recordings exhibit remarkable dynamic fidelity and spatial nuance, demonstrating the viability of non-contact media archaeology.

REFERENCES & PRIMARY CITATIONS

- Fadeyev, V. & Haber, C. (2003). Reconstruction of mechanically recorded sound by image processing. JAES.
- Takahashi, Y. (2023). Optical Recovery of Grooved Historical Media. ZIAA Press.
- Kittler, F. (1999). Gramophone, Film, Typewriter. Stanford University Press.

Archival specimen — ZIAA Publications Division • <https://zazieinstitute.org/monographs/essay-2023-02> • Permanent preservation record — Cite as: Dr. Yuna Takahashi et al., "Media Archaeology of Inscribed Sound: Optical Reconstruction and Artifact Isolation in Fragile Historical Recordings," Volume V, Issue 2 (Archival Audio & Signal Archaeology), ZIAA Press, 2023.